batmonitor 0.52-c: one cycle (bm_step, every 5 s) | blue = pure decision function (bits + state, enumerated by make check) | green = discharge matrix | [CBMC] = proved for all inputs

inputs (chain_inputs)
PCC, Sofar Bat1 W + SoC, em0 WP W, R290 Hz, PV,
BMS SoC / U / CCL / DCL / lowest cell (0.52-c), season, forecast
-> stale, wp_fresh, wp_on (≥ WP_ON_TH), night (PV < NIGHT_PV_TH)

D1 bm_d1 state: fc_active, bat1_first
FC_ACTIVE: lowest SoC > target + FC_HYST (off < target)
FC_BAD: target ≥ FC_BAD_TARGET
B1FIRST: Bat1 SoC > BAT1_SOC_MIN (on from +2 %) and WP on
[CBMC h_d1_bits]

wp_kind

src

S2 heat pump kind
forecast cap binds -> **BAD**
else season_wp (FC_ACTIVE = summer):
winter -> **NONE**, transition -> **CAP**, summer -> **ALL**

S3 source (measured season only, 0.45-c)
day -> **DAY**
night + B1FIRST -> **B1**
night, not (winter + WP on) -> **STK** (night floor)
night, winter + WP on -> **B1W**

S0 gates per bank (chain_gates)
SoC from the BMS; caps per phase, one per bank:
chg_cap = min(charger x U, BMS CCL) dis_cap = min(DISCHARGE_MAX_PHASE, BMS DCL)

wp_share
NONE 0 | CAP min(WP, WP_BAT_MAX_TRANSITION) | ALL WP | BAD WP x WP_BAT_SHARE_BAD %
z2 = PCC + (WP - share) (the point the stacks regulate on)

z2 surplus

D2 bm_d2 per bank state: prot, force, low, chg_prev
first match: BMS_MISSING > FORCE (force or low) > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > **DIS**
LOW (0.52-c, state low): on at DCL < DCL_FORCE_A 1 A or lowest cell < CELL_FORCE_V 3.00 V, off when DCL is back and the cell ≥ CELL_FORCE_RELEASE_V 3.25 V; LOW -> FORCE (charge, no discharge)
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5184 · STALE 1440 · LADESPERRE 360 · PROT 324 · CHARGE 288 · CHARGING 198 · DIS 198 · FULL 72
[CBMC h_d2_bits; threshold order needed: h_d2_bits_any_order]

row

D-lead bm_dlead state: lead
|SoC1 - SoC2| > SOC_BALANCE_ON -> the stack ahead leads
< SOC_BALANCE_OFF -> no lead

z2

wp_cap_armed
(em0 stale, winter, R290 on)

mode per phase

DIS

else

CHARGE

discharge matrix (0.45-c, whole watts)

BM_SRC_RULE[src]

SRC	b1_signed	trickle	idle_w	prop_hold
DAY	0	0	20	0
B1	0	0	20	0
STK	1	1	0	1
B1W	0	0	20	0

dis_cap

lead

BMS_MISSING / FORCE / STALE / LADESPERRE / FULL / PROT / CHARGING
-> fixed setpoint (0, force charge incl. LOW, ...)

deficit (int W)
b1 = b1_signed ? Bat1 : max(Bat1 charge, 0)
target = SOYO_TARGET + (trickle ? SOFAR_TRICKLE : 0)
deficit = own_discharge - b1 - KP_PM x (z2 + own_charge - target) / 1000

charge (chain_charge)
want = max(0, own_charge + KP x surplus)
alloc_charge: CHARGE_PRIORITY, lead stack last, up to chg_cap
[CBMC h_charge]

D3 bm_d3 state: prop_prev
PROP = import (< -100 W) or
((prop_prev or prop_hold) and deficit > 20 W)

dis_amount: w = min(PROP ? max(deficit, 0) : idle_w, W_MAX, WP_CAP if armed)
[CBMC h_dis_amount]

alloc_discharge on **alloc_levels** (0.48-c: one split core for charge and discharge)
lead: level 0 = lead stack (BAL), level 1 = only what its caps cut (SPILL / WAIT)
no lead: one level, every stack the same share (EQ)
within a level: water filling up to dis_cap
[CBMC h_alloc_discharge: within caps, total ≤ w, lead first, equal per stack]

setpoints per phase -> Multis (ESS AcPowerSetpoint)
rule text: <mode> WP:<reason> SRC:<src> <split> [WPCAP] [CCL/DCL]
[CBMC h_discharge: whole discharge path]

PI prototype (shadow, not armed)